230-231页2,3,4,6,7,8.
241-243页2,3,5,6,19,20.

f_T denotes the characteristic polynomial, and p_T the minimal polynomial.

230-2

Let $T \in \mathcal{L}(\mathbb{R}^3)$ represented by $\mathrm{diag}(2, 2, -1)$. Prove that T has no cyclic vector. What is the T-cyclic subspace generated by the vector $(1, -1, 3)$?
Proof: Note that $p_T = (x-2)(x+1)$ has degree 2 so T has no cyclic vector. For $v = (1, -1, 3)$, $Tv = (2, -2, -3)$, so v, Tv are linearly independent. Since $(T^2 - T - 2I)v = 0$, the cyclic subspace is generated by v, Tv, which is $\{(x, y, z) : x + y = 0\}$.

230-3

Let $T \in \mathcal{L}(\mathbb{C}^3)$ represented by

$$\begin{pmatrix} 1 & i & 0 \\ -1 & 2 & -i \\ 0 & 1 & 1 \end{pmatrix}.$$

Find the $T-$annihilator of the vectors $(1, 0, 0)$ and $(1, 0, i)$.
Solution: $f_T = p_T = (x-1)(x^2 - 3x + 2 + 2i)$.
For $v = (1, 0, 0)$, $Tv = (1, -1, 0)$ and $T^2 v = (1 - i, -3, -1)$ are linearly independent, so the annihilator is $p_T = (x-1)(x^2 - 3x + 2 + 2i)$.
For $v = (1, 0, i)$, $Tv = v$ so the annihilator is $x - 1$.

230-4

Prove that if T^2 has a cyclic vector, then T has a cyclic vector. Is the converse true?
Proof: If T^2 has a cyclic vector v, then $v, T^2 v, T^4 v, \cdots$ spans V. Hence $v, Tv, \cdots$ spans V, so T has a cyclic vector v.
The converse is false: Let $V = \mathbb{R}^2$, $T : (x, y) \mapsto (y, x)$. Then T has a cyclic vector $(1, 0)$ but $T^2 = id$ has no cyclic vector.

230-6

Give a direct proof that if A is the companion matrix of the monic polynomial p, then $p_A = p$.
Proof: Clearly $e_1, Ae_1, \cdots, A^{n-1} e_1$ are linearly independent, so $\deg p_A = n$ and $p_A = f_A$. Hence $p_A = f_A = p$.

230-7

Let V be an $n-$dimensional vector space, $T \in \mathcal{L}(V)$. Suppose T is diagonalizable.
(a) If T has a cyclic vector, show that T has n distinct eigenvalues. (b) If T has n distinct eigenvalues, and $\alpha_1, \cdots, \alpha_n$ is a basis of eigenvectors of T, then $\alpha = \alpha_1 + \cdots + \alpha_n$ is a cyclic vector for T.
Proof: (a) Note that $\deg p_T$ is the number of distinct eigenvalues, and T is cyclic implies $\deg p_T = n$.
(b) Suppose $T\alpha_i = c_i\alpha_i$, then under this basis the n vectors $T^k\alpha$ form the Vandermonde matrix, hence it has non-zero determinant, so $T^k\alpha$ form a basis of V and α is a cyclic vector for T.

230-8

Let $T \in \mathcal{L}(V)$ with V finite dimensional. Suppose T has a cyclic vector v. Prove that if $U \in \mathcal{L}(V)$ commutes with T, then U is a polynomial in T.
Proof: Suppose $Uv = c_0v + c_1Tv + \cdots + c_{n-1}T^{n-1}v$, we show that $U = p(T) = c_0I + \cdots + c_{n-1}T^{n-1}$:
For any $k \geqslant 0$, $UT^kv = T^kUv = T^kp(T)v = p(T)T^kv$. Since T^kv span V, $U = p(T)$ is a polynomial in T.

241-2

Let $T \in \mathcal{L}(V)$, $R = \mathrm{Im}T$.
(a) Prove that R has a complementary $T-$invariant subspace iff R is independent of $N = \mathrm{Ker}T$.
(b) If R, N are independent, prove that N is the unique $T-$invariant subspace complementary to R.
Proof: (a)(b) If $R \cap N = 0$, then $\dim(R+N) = \dim R + \dim N = \dim V$ so $R \oplus N = V$. Since N is $T-$invariant, R has a complementary $T-$invariant subspace.
If R has a complementary $T-$invariant subspace W, then $\dim W = \dim N$. Note that $T(W) \subset W$ and $T(W) \subset R$, so $T(W) \subset R \cap W = 0$. Hence $W \subset N$ so $W = N$, and R is independent of N.

242-3

Let $T \in \mathcal{L}(\mathbb{R}^3)$ represented by

$$\begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Let $W = \mathrm{Ker}(T - 2I)$. Prove that W has no complementary $T-$invariant subspace.
Proof: $W = \langle (0,1,0)^T \rangle$. If $\mathbb{R}^3 = W \oplus V$ where V is $T-$invariant, $f_T = (x-2)^2(x-3) = f_W f_T$. Hence $f_T = (x-2)(x-3)$ so 2 is an eigenvalue of $T|_V$, a contradiction to $W = \mathrm{Ker}(T-2I)$.

242-5

Let $T \in \mathcal{L}(V)$. If $f \in F[x]$ and $\alpha \in V$, let $f\alpha = f(T)\alpha$. If $V_1, \cdots, V_k$ are $T-$invariant subspaces and $V = V_1 \oplus \cdots \oplus V_k$, show that $fV = fV_1 \oplus \cdots \oplus fV_k$.
Proof: Since V_i are $T-$invariant, $fV_i \subset V_i$ so they are independent. For $f\alpha = f(T)\alpha \in fV$, suppose $\alpha = \alpha_1 + \cdots + \alpha_k$ where $\alpha_i \in V_i$, then $f\alpha = f\alpha_1 + \cdots + f\alpha_k \in fV_1 + \cdots + fV_k$. Hence $fV = fV_1 \oplus \cdots \oplus fV_k$.

242-6

Following Ex5, suppose $\alpha, \beta \in V$ have the same $T-$annihilator. Prove that, for any polynomial f, $f\alpha$ and $f\beta$ have the same annihilator.
Proof: Suppose the annihilator of α, β is g. Then $h(f\alpha) = 0 \iff (hf)\alpha = 0 \iff g \mid hf$, hence the annihilators of $f\alpha, f\beta$ are both $g/\gcd(g, f)$.

243-19

Let $T \in \mathcal{L}(V)$ with $\dim V < \infty$. Prove that T has a cyclic vector iff the following is true: every $U \in \mathcal{L}(V)$ which commutes with T is a polynomial in T.
Proof: The only if part is proved in Ex230-8.
Conversely, if T has no cyclic vector, then it admits a cyclic decomposition $V = \oplus R\alpha_i$. Consider $U \in \mathcal{L}(V)$ such that $U|_{R\alpha_1} = I_{R\alpha_1}$ and $U|_{R\alpha_i} = 0 \forall i > 1$. Then U commutes with T but it is not a polynomial in T: If $U = p(T)$, then $p \equiv 1 \pmod{p_1}$ but $p \equiv 0 \pmod{p_2}$, a contradiction with $p_2 \mid p_1$.

243-20

Let $T \in \mathcal{L}(V)$ with $\dim V < \infty$. We ask when it is true that every non-zero vector in V is a cyclic vector for T. Prove that this is the case iff f_T is irreducible over F.
Proof: If f_T is irreducible over F, then for every $\alpha \in V \backslash \{0\}$, if $R\alpha \neq V$, then $R\alpha$ is a $T-$invariant subspace, hence $f_T = f_{R\alpha} \cdot f_{V/R\alpha}$ is not irreducible. Hence every non-zero vector in V is a cyclic vector for T.
Conversely, if $f_T = g \cdot h$ with non constant g, h, then $W = \mathrm{Ker} g(T) \neq 0$ is a $T-$invariant proper subspace. Take $\alpha \in W \backslash \{0\}$, then $\{\alpha, T\alpha, \cdots, T^{n-1}\alpha\}$ can only span W, so α is not a cyclic vector.